Velocities that follow Larson's size–linewidth relation in the HENDOS fractal generator

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1 Overview

The HENDOS initial conditions generator `makeics` can make a box fractal of stars (Goodwin & Whitworth 2004) and give the stars velocities in one of four ways. One of them, `larson` (made by the routine `larsonfractal`), is designed so that the velocity difference between two stars grows as the square root of their separation,

$$\delta v \propto d^{1/2}, \tag{1}$$

which is the slope of the size–linewidth relation of molecular clouds. This note explains how that is done, shows that the result follows equation (1) exactly on average, describes how the velocities are put on the observed relation in $\mathrm{km\,s^{-1}}$, and compares the other three velocity options.

The short answer: `larson` gives the right *slope* at every scale by construction, and the *normalisation* is set to the observed relation of Solomon et al. (1987). Unlike every other option in the generator, the velocities are therefore not scaled to a virial ratio: the virial ratio is an outcome. The other options do not give this slope.

2 The size–linewidth relation

Larson (1981) found that the velocity dispersion σ of molecular clouds and of regions within them increases with their size L as

$$\sigma \simeq 1.10\,\mathrm{km\,s^{-1}} \left(\frac{L}{\mathrm{pc}}\right)^{0.38}. \tag{2}$$

Later work with more uniform data found a steeper slope; the relation most often quoted is that of Solomon et al. (1987),

$$\sigma \simeq 0.72\,\mathrm{km\,s^{-1}} \left(\frac{R}{\mathrm{pc}}\right)^{0.5}, \tag{3}$$

where σ is the one-dimensional velocity dispersion and R the radius of the cloud. Heyer et al. (2009) showed that the coefficient is not universal but increases with the surface density of the cloud. The generator aims at the slope $1/2$ of equation (3).

The relation is a statement about gas. Using it for stars assumes that stars form with the velocities of the gas they form from, so that stars that form close together have similar velocities, and stars that form far apart differ by more.

3 The method

3.1 A Gaussian velocity field with a linear covariance

Once the N star positions $\mathbf{x}_i$ are fixed, each Cartesian component of the velocities is drawn as a set of N Gaussian random numbers with zero mean and covariance

$$\langle v_{i,a}\, v_{j,b}\rangle = \delta_{ab}\, C_{ij}, \qquad C_{ij} = v_{\rm s}^2 \left(D - \tfrac{1}{2} d_{ij}\right), \tag{4}$$

where a, b label the components, $d_{ij} = |\mathbf{x}_i - \mathbf{x}_j|$ is the separation of stars i and j, $v_{\rm s}$ is a velocity scale (`vscale`, with units of velocity per square root of length) and D is a constant length. The code uses $D = 10\, d_{\rm max}$, ten times the largest separation in the system. The three components are independent of each other.

3.2 Why this gives the square-root law

The mean square difference between one component of the velocities of two stars is

$$\left\langle (v_{i,a} - v_{j,a})^2 \right\rangle = C_{ii} + C_{jj} - 2C_{ij} = v_{\rm s}^2 D + v_{\rm s}^2 D - 2v_{\rm s}^2 \left(D - \tfrac{1}{2} d_{ij}\right) = v_{\rm s}^2\, d_{ij}. \tag{5}$$

The constant D cancels, and what is left is exactly proportional to the separation. Summing the three components,

$$\left\langle |\mathbf{v}_i - \mathbf{v}_j|^2 \right\rangle = 3\, v_{\rm s}^2\, d_{ij}, \qquad \text{so} \qquad \delta v_{\rm rms}(d) = \sqrt{3}\, v_{\rm s}\, d^{1/2}. \tag{6}$$

This holds for every pair of stars, at every separation, whatever the positions are. It does not depend on the fractal dimension.

3.3 The velocity dispersion of a region

Larson's relation is usually stated for the velocity dispersion of a region rather than for pairs. The two are connected by an identity: for any set S of n stars with mean velocity $\bar{\mathbf{v}}_S$,

$$\sigma_S^2 \equiv \frac{1}{n} \sum_{i \in S} |\mathbf{v}_i - \bar{\mathbf{v}}_S|^2 = \frac{1}{2n^2} \sum_{i,j \in S} |\mathbf{v}_i - \mathbf{v}_j|^2. \tag{7}$$

Taking the average with equation (6),

$$\left\langle \sigma_S^2 \right\rangle = \tfrac{3}{2}\, v_{\rm s}^2\, \bar{d}_S, \qquad \left\langle \sigma_{S,1{\rm D}}^2 \right\rangle = \tfrac{1}{2}\, v_{\rm s}^2\, \bar{d}_S, \tag{8}$$

where $\bar{d}_S$ is the mean separation of pairs of stars in the region. The mean separation is proportional to the size of the region, so $\sigma \propto L^{1/2}$. For example, the mean distance between two points in a uniform sphere of radius R is $36R/35$, so for stars spread uniformly through such a sphere

$$\sigma_{1{\rm D}} = \left(\frac{18}{35}\right)^{1/2} v_{\rm s}\, R^{1/2} \simeq 0.72\, v_{\rm s}\, R^{1/2}. \tag{9}$$

By coincidence the numerical factor is the same as the coefficient in equation (3): a velocity scale $v_{\rm s} = 1\,{\rm km\,s^{-1}\,pc^{-1/2}}$ corresponds to the Solomon et al. relation for a uniformly filled region.

3.4 The constant length and the bulk velocity

Equation (4) gives every star a velocity variance $v_{\rm s}^2 D$ per component, which is much larger than the differences between stars. Almost all of it is a single velocity shared by every star: the correlation between any two stars is at least $1 - d_{\rm max}/2D = 0.95$. This shared velocity carries

no information and is removed when the generator moves the stars to their centre-of-velocity frame. What remains are the differences, which obey equation (6).

D is needed to make equation (4) a valid covariance, that is, to make the matrix C positive definite so that Gaussian numbers with that covariance exist. For weights w_i,

$$\sum_{i,j} w_i C_{ij} w_j = v_{\rm s}^2 D \Big(\sum_i w_i \Big)^2 - \tfrac{1}{2} v_{\rm s}^2 \sum_{i,j} w_i \, d_{ij} \, w_j . \tag{10}$$

For distinct points in Euclidean space the second sum is negative whenever $\sum_i w_i = 0$ (the distance $|\mathbf{x} - \mathbf{y}|$ is conditionally negative definite; Schoenberg 1938), so the form is positive in those directions, and a large enough D makes it positive in the remaining direction. In the tests of Section 7 the smallest eigenvalue of C was positive in every case with $D = 10\, d_{\rm max}$.

3.5 Relation to turbulent velocity fields

A Gaussian field whose increments have variance proportional to d^{2H} is a fractional Brownian field with Hurst exponent H. Equation (6) is the case $H = 1/2$, for which the three-dimensional power spectrum of the velocity is $P(k) \propto k^{-(3+2H)} = k^{-4}$. This is the spectrum commonly used to set up turbulent velocity fields in simulations of star formation, and is that of Burgers (shock-dominated) turbulence. The method is therefore equivalent to sampling such a turbulent field at the positions of the stars, without having to make the field on a grid. Because the three components are independent, the field is a mixture of solenoidal and compressive motions in the natural ratio $2 : 1$.

4 Implementation

The routine is `larsonfractal` in `HENDOS/src/ics/fractalsubs.f90`. It

1. calls the routine `makefractal` to make the positions and cut the number of stars to N;
2. finds the largest separation $d_{\rm max}$ and sets $D = 10\, d_{\rm max}$;
3. fills the $N \times N$ matrix C of equation (4);
4. factorises it as $C = \mathsf{L}\,\Delta\,\mathsf{L}^{\mathsf{T}}$, with L unit lower triangular and Δ diagonal (a Cholesky decomposition without square roots);
5. for each component draws N independent Gaussian numbers z_k of unit variance and sets

$$v_i = \sum_k \mathsf{L}_{ik} \, \Delta_k^{1/2} \, z_k , \tag{11}$$

   which has covariance $\mathsf{L}\Delta\mathsf{L}^{\mathsf{T}} = C$ as required;
6. subtracts the mean velocity.

The Gaussian numbers come from `gasdev`, which uses the random number generator `ran3`.

Cost. The matrix needs $8N^2$ bytes (8 MB for $N = 10^3$, 0.8 GB for $N = 10^4$) and the factorisation takes of order $N^3/3$ operations. The method is practical for a few thousand stars and becomes slow beyond about 10^4.

5 Normalisation

For every other kind of initial conditions `makeics` multiplies all the velocities by one factor to give the virial ratio the user asks for. A Larson fractal is the exception: once the stars have their masses and the fractal has its size, the velocities are set to the observed relation in $\mathrm{km\,s^{-1}}$, and the virial ratio `qvir` in the input is not used.

5.1 Setting the coefficient

The routine `larsonfractal` is called with radius 1 and $v_\mathrm{s} = 1$, so that by equation (6) the mean square difference of one velocity component between two stars equals their separation in units of the radius. The positions are then multiplied by the radius R (in pc) and the velocities by

$$v_\mathrm{s}\, R^{1/2}, \qquad v_\mathrm{s} = \texttt{vlarson}, \tag{12}$$

which gives, with d in pc,

$$\left\langle (v_{i,a} - v_{j,a})^2 \right\rangle = v_\mathrm{s}^2\, d \quad (\mathrm{km\,s^{-1}})^2. \tag{13}$$

The input `vlarson` is therefore the rms difference, in $\mathrm{km\,s^{-1}}$, of one velocity component between two stars 1 pc apart. Its standard value is $1\,\mathrm{km\,s^{-1}}$, for which equation (8) gives a one-dimensional velocity dispersion of $0.72\,(R/\mathrm{pc})^{1/2}\,\mathrm{km\,s^{-1}}$ for a region filled uniformly to radius R: the relation of Solomon et al. (1987), equation (3). Other coefficients (for example those that depend on surface density; Heyer et al. 2009) are obtained by changing `vlarson`. Larson's (1981) own relation has a different slope and cannot be matched in this way.

A fractal is not uniformly filled, so the dispersion of the whole system is not exactly $0.72\,v_\mathrm{s} R^{1/2}$: it is $(\bar{d}/2)^{1/2} v_\mathrm{s}$, where $\bar{d}$ is the mean separation of the stars, which is smaller for a clumpier fractal. What is fixed is the relation between pairs of stars.

5.2 The virial ratio that results

From equation (8) applied to the whole system of mass M, the kinetic energy is $T = \frac{3}{4} M v_\mathrm{s}^2 \bar{d}$ on average, so the virial ratio is

$$Q = \frac{T}{|\Omega|} = \frac{3\,M\,v_\mathrm{s}^2\,\bar{d}}{4\,|\Omega|}. \tag{14}$$

For a uniform sphere ($|\Omega| = 3GM^2/5R$, $\bar{d} = 36R/35$) this is

$$Q = \frac{9}{7}\,\frac{v_\mathrm{s}^2 R^2}{GM} \simeq 0.30 \left(\frac{v_\mathrm{s}}{1\,\mathrm{km\,s^{-1}}}\right)^2 \left(\frac{R}{\mathrm{pc}}\right)^2 \left(\frac{M}{1000\,\mathrm{M_\odot}}\right)^{-1}. \tag{15}$$

The virial ratio rises as R^2/M: large, low-mass systems are unbound and small, massive ones are cold. A clumpy fractal has a larger $|\Omega|$ and a smaller $\bar{d}$ than a uniform sphere of the same mass and radius, so a smaller Q. Table 1 gives the values found for the standard mass function. `makeics` prints the virial ratio of the system it has made and records it in the file.

R (pc)	M ($\mathrm{M_\odot}$)	Q	σ_1D ($\mathrm{km\,s^{-1}}$)
0.5	377	0.10 ± 0.04	0.46
1.0	368	0.43 ± 0.16	0.68
2.0	375	1.54 ± 0.80	0.95

Table 1: Virial ratio Q and one-dimensional velocity dispersion of Larson fractals of $N = 1000$ stars, fractal dimension 1.6, with `vlarson` $= 1\,\mathrm{km\,s^{-1}}$ and masses from the standard mass function. Means over six fractals; the error is the scatter between fractals.

The scatter in Q between fractals is large, because both $|\Omega|$ and the few large-scale motions that carry most of the kinetic energy vary from one fractal to the next. If a particular virial ratio is needed as well as the Larson slope, the radius or `vlarson` can be adjusted until the printed value is the one wanted.

6 The other velocity options

The generator has three other velocity fields, selected by `velocities` in the input file and made by the routine `makefractal` (the `vtype` given below is its argument). For these the velocities are scaled to the virial ratio asked for. They are built into the fractal as it is made: at each level of the hierarchy a box (parent) is divided into eight (with the standard two boxes per side) and some of the children survive. Let δ_ℓ be the half-width of a box at level ℓ, which halves at each level.

`inherit` (`vtype='f'`). Each child takes its parent's velocity plus a Gaussian random velocity of dispersion $v_\mathrm{s}\,\delta_\ell$ per component. Two stars whose last common ancestor is at level ℓ share all the kicks down to that level and differ by those below it, so

$$\left\langle |\mathbf{v}_i - \mathbf{v}_j|^2 \right\rangle = 6\,v_\mathrm{s}^2 \sum_{k>\ell} \delta_k^2 = 6\,v_\mathrm{s}^2 \delta_\ell^2 \sum_{m\geq 1} 4^{-m} = 2\,v_\mathrm{s}^2 \delta_\ell^2 . \tag{16}$$

Stars that separate at level ℓ are typically a box width apart, so this is $\delta v \propto d$, a slope of 1 rather than 1/2. The measured slope is somewhat less than 1 (Table 2) because separation is only loosely tied to the level: stars in neighbouring boxes can be close together in space but have a distant common ancestor. A slope of 1/2 could be made this way by using kicks proportional to $\delta_\ell^{1/2}$; this is not implemented.

`standard` (`vtype='s'`). The velocities of the original `makefractal`: each child takes its parent's velocity plus a Gaussian random velocity of the *same* dispersion at every level. Then $\left\langle |\mathbf{v}_i - \mathbf{v}_j|^2 \right\rangle \propto \ell_\mathrm{max} - \ell$, the number of levels below the common ancestor, which grows only as the logarithm of the separation. Nearby stars do have similar velocities, but there is no power law, and a fitted slope is small.

`random` (`vtype='u'`). Every star has an independent Gaussian velocity. The velocity difference does not depend on separation.

7 Tests

The tests were made with a line-by-line Python copy of the routine, not with the Fortran, which had not been compiled when this was written. Fractals of $N = 1000$ stars with fractal dimension 1.6 and radius 1 were made, the rms velocity difference of all pairs of stars was measured in 12 logarithmic bins of separation (from about 0.05 to 1.5), and a power law was fitted.

Four further checks were made of `larson`:

- Drawing 200 sets of velocities for one set of positions, the ratio $\left\langle |\mathbf{v}_i - \mathbf{v}_j|^2 \right\rangle / 3v_\mathrm{s}^2 d$ was between 0.99 and 1.04 in every bin, confirming equation (6).
- The matrix C was positive definite for every fractal made.
- For fractal dimensions 2.0 and 2.6 the slopes of single fractals were 0.46±0.05 and 0.44±0.05 (five fractals each).

velocities	vtype	slope, coresize $= 0$	slope, coresize $= 0.003$
larson	–	0.50 ± 0.07	0.50 ± 0.08
inherit	f	0.80 ± 0.09	0.79 ± 0.12
standard	s	0.12 ± 0.02	0.10 ± 0.02
random	u	0.00 ± 0.00	0.00 ± 0.00

Table 2: Slope of the rms velocity difference between stars against their separation. Each entry is the mean over ten fractals; the error is the scatter between fractals, not the error on the mean.

- With the velocities set as in Section 5 and vlarson $= 1\,\mathrm{km\,s^{-1}}$, the coefficient measured from the pairs of stars was 1.02 to 1.04, and uniform spheres (fractal dimension 3) of radius 1 pc had $\sigma_{1\mathrm{D}} = 0.71\,\mathrm{km\,s^{-1}}$, against the 0.72 expected.

The scatter between fractals is real and is worth remembering: any one set of initial conditions is a single draw, and its largest scales are sampled by only a few independent motions, so its measured slope can differ from 1/2 by about 0.1. Fitting the logarithm of a noisy average also biases the slope of a single fractal slightly low. The relation is exact only on average.

8 Limitations

- The virial ratio is not controlled: it follows from the mass and size (Section 5) and varies a good deal between fractals.
- The observed coefficient is not universal (Heyer et al. 2009); the standard value is that of Solomon et al. (1987).
- The velocities are Gaussian. Real turbulence is intermittent, with non-Gaussian tails, and has correlations between velocity and density that are absent here: the velocity field knows about the positions of the stars only through their separations.
- The observed relation is for gas, measured from line widths along the line of sight within projected regions. The generator imposes the equivalent relation between stars in three dimensions.
- Memory and time limit the method to about 10^4 stars.

9 Input file

The parameter file ics.nml for a Larson fractal:

```
&ics
  outfile = 'data/frac1k.dat'
  n = 1000
  seed = 12345
  distribution = 'fractal'
  radius = 1.0 ! total radius (pc)
/
&fractal
  fdim = 1.6
  velocities = 'larson'
  vlarson = 1.0 ! km/s at 1 pc
/
```

Run it in HENDOS/run with ./makeics ics.nml. All the inputs are described in the file PARAMS_README.md in that directory.

References

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- Schoenberg I. J., 1938, Trans. Amer. Math. Soc., 44, 522
- Solomon P. M., Rivolo A. R., Barrett J., Yahil A., 1987, ApJ, 319, 730